

KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Computer Science and Engineering
Spring-2023 Mid-term Examination
Program: B.Sc. in CSE (Batch – 14th) 1st Year 2nd Semester
Course Title: Physics - II
Course Code: PHY 1201

PART-A: MCQ

Time: 10 Minutes

Total Marks: 10x1=10

Student's ID:

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Date:

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 Invigilator's Signature:

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(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

Cognitive Understanding

- What is the relationship between the energy of an electron and its orbit radius in the Bohr atomic model?
A) Directly proportional
B) Inversely proportional
C) Proportional to the square of its radius
D) Inversely proportional of its radius
E) Not related

Cognitive Understanding

- For elastic collision, which is the correct statement between kinetic energy before (K_b) and after (K_a) collision?
A) $K_b > K_a$
B) $K_b < K_a$
C) $K_b > 0$ & $K_a = 0$
D) $K_b = 0$ & $K_a > 0$
E) $K_b = K_a$

Cognitive Knowledge

- What is the unit of mechanical equivalent of heat?
A) calJ^{-1}
B) Jcal^{-1}
C) JK^{-1}
D) KJ^{-1}
E) calJ

Cognitive Application

- The area of the curve between which two quantities indicates the amount of work done? (Where, P -Pressure, V -Volume, S -Entropy, T -temperature, W -external work done)
A) P & T
B) P & S
C) P & W
D) P & V
E) V & T

Cognitive Knowledge

- Which of the following change is only possible in slower process?
A) Isothermal
B) Isochoric
C) Isobaric
D) Adiabatic
E) Iso-entropic

Cognitive Knowledge

6. How many adiabatic processes are available in the Carnot cycle?
- A) 0
 - B) 1
 - C) 2
 - D) 3
 - E) 4

Cognitive Application

7. If the mean free path of atoms is doubled, then what will be the pressure of the gas?
- A) $4P$
 - B) $2P$
 - C) P
 - D) $3P/2$
 - E) $P/2$

Cognitive Understanding

8. Which of the following spectral series will be produced when the electron transition occurs from all higher orbits to the third orbit?
- A) Lyman
 - B) Balmer
 - C) Paschen
 - D) Brackett
 - E) Pfund

Cognitive Application

9. Which of the following equation represents the change in entropy for a reversible process at constant temperature?
- A) $dS = \frac{dQ}{T}$
 - B) $dS = m \ln\left(\frac{T_2}{T_1}\right)$
 - C) $dS = m \ln\left(\frac{V_2}{V_1}\right)$
 - D) $dS = \frac{ml_f}{T}$
 - E) $dS = 0$

Cognitive Knowledge

10. What is the energy of the 1st orbit of hydrogen atom?
- A) -17.6 eV
 - B) -13.6 eV
 - C) -1.5 eV
 - D) -0.53 eV
 - E) 0.53 eV

Signature of Examiner

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PART-B: SAQ

Time: 1 Hour and 20 Minutes

Total Marks: 4x5=20

INSTRUCTIONS: Answer question no. 01 (one) and any 03 (three) from the rest. Marks are shown by side.

1. See the table given below and answer the questions that follows:

Engine	Source		Sink	
	Temperature °C	Absorbed Heat (J)	Temperature °C	Rejected Heat (J)
Engine-I	327	X	27	600
Engine-II	527	1800	127	900

- a) What are the conditions for increasing the efficiency of a heat engine? [1]
b) Find out the value of X. [1]
c) Which engine is better? Explain your response. [3]
2. Read the stem given below and answer the question that follows:
“For a gas $\gamma = 1.66$ and $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$ ”
- a) State Mayer Hypothesis. [1]
b) Establish the relationship between two constants mentioned in the stem. [1]
c) Calculate the values of molar specific heats of the gas. [3]
3. a) How is pedesis in the fluid caused? [1]
b) What are the factors that affect the Mean Free Path? [1]
c) The diameter of the molecule of a gas is $3 \times 10^{-10} \text{ m}$ and number of molecules per cm^3 is 6×10^{20} . Calculate the mean free path of the molecules. [3]
4. a) State the Zeroth law of thermodynamics. [1]
b) Illustrate the pressure – volume curve for adiabatic and isothermal process. [2]
c) Comments about the far future of universe related to entropy with logical explanation. [2]
5. a) What is Bohr formula for energy? [1]
b) What is the radius of the first Bohr orbit in Li^{2+} ? [2]
c) How does the Bohr atom model relate to the quantization of energy? [2]